1. Introduction



Motivation

- Income prediction supports policy, finance, and fairness.
- Income inequality is one of the most worrying challenges.
- We knew that income distribution is not fair.

ML vs Statistics

- ML offers adaptability over statistics.
- ML can identify complex patterns.
- ML can achieve amazing performance.

We focused on real world scenarios

- High Risk Loan Approval
- Social Security Measures
- IRS auditing.

2. Data Description

Dataset Shape (48843, 14)

- age, **race** , **sex**, **native-country** , **education** , education-num, **workclass** , **occupation** , hours-per-week, capital-gain, capital-loss, **marital-status** , **relationship** , fnlwgt

Predict

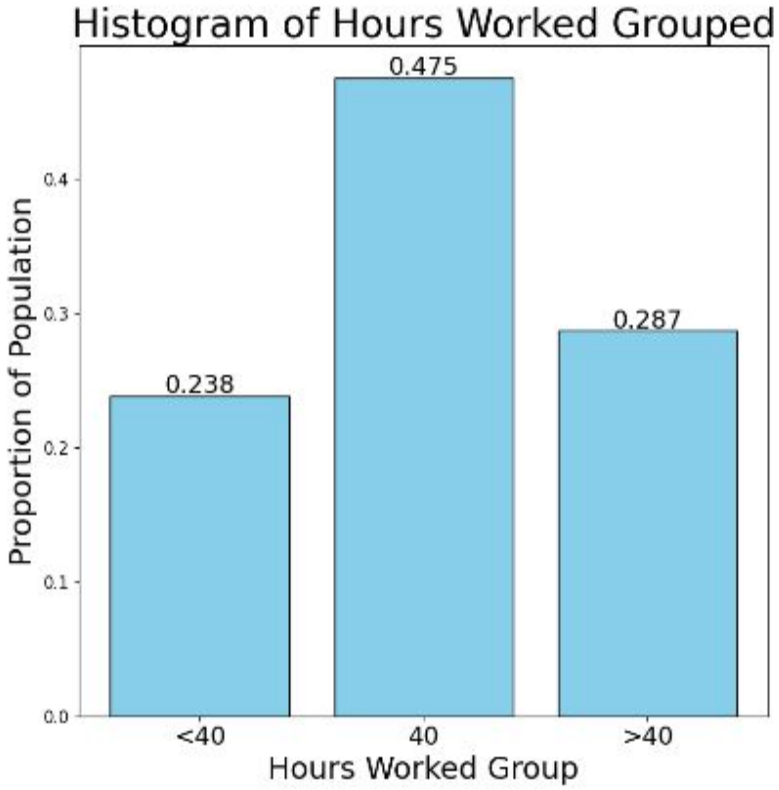
(>50k) or (<=50k)

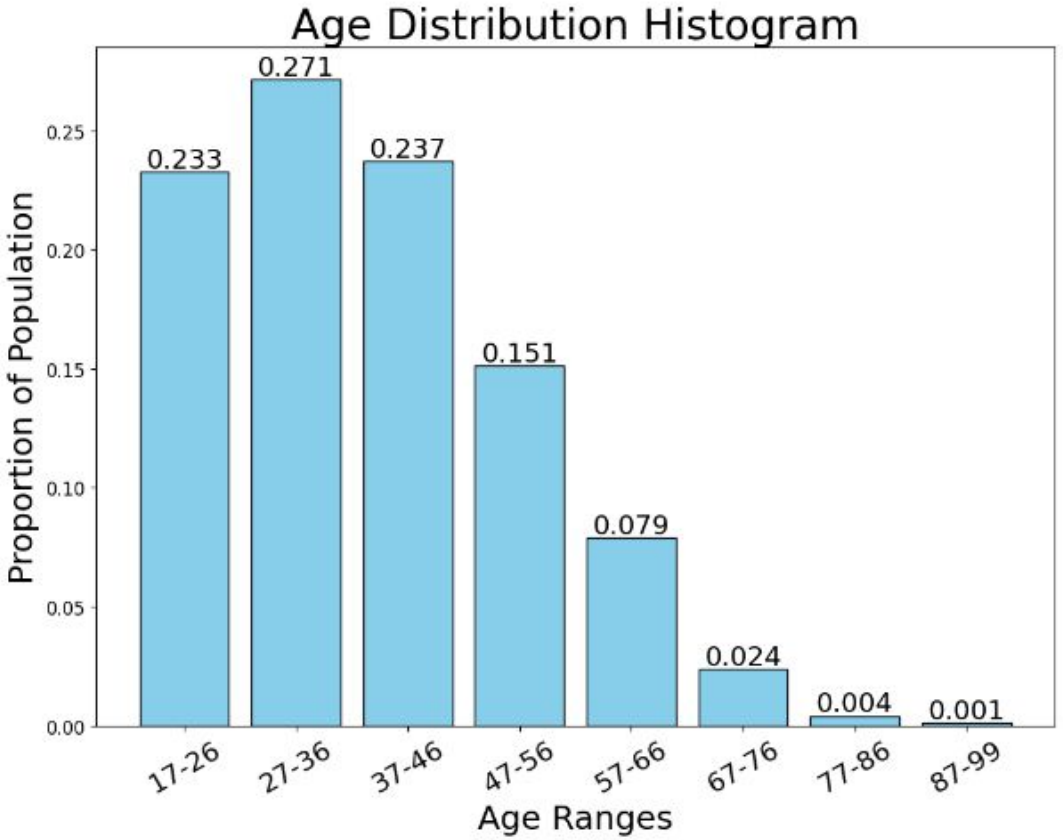
- 8 categorical features
- 6 numerical features (1 discrete 5 continuous)

Problems

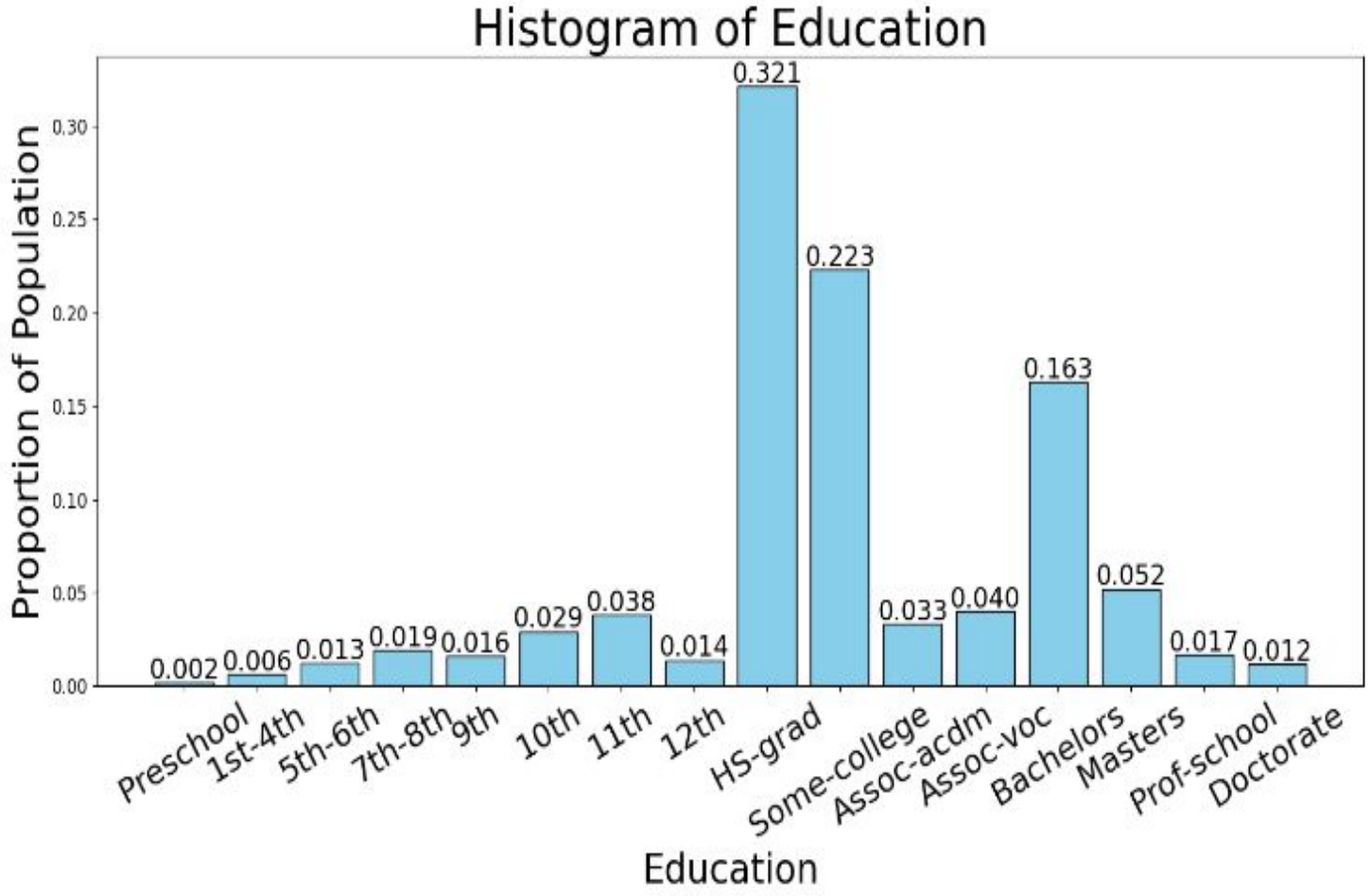
- **Class Imbalance:** 24% (>50k)
- **Missing Values:** workclass, occupation, native-country
- **Demographic Imbalance:** (85.7% White) and (67% Male)
- **Skewed Distributions:** capital-gain/loss, hours-per-week
- **Feature Correlation:** education, education-num

3. Data Visualization





3. Data Visualization



4. Dataset Preprocessing

Feature	Most Common Value	% Frequency
workclass	Private	73.64%
occupation	Prof-specialty	13.41%
native-country	United-States	91.35%

Mode input for missing values, to avoid losing 7.41% of our dataset.

One-Hot Encoded Features
workclass
marital-status
occupation
relationship
race
sex
native-country

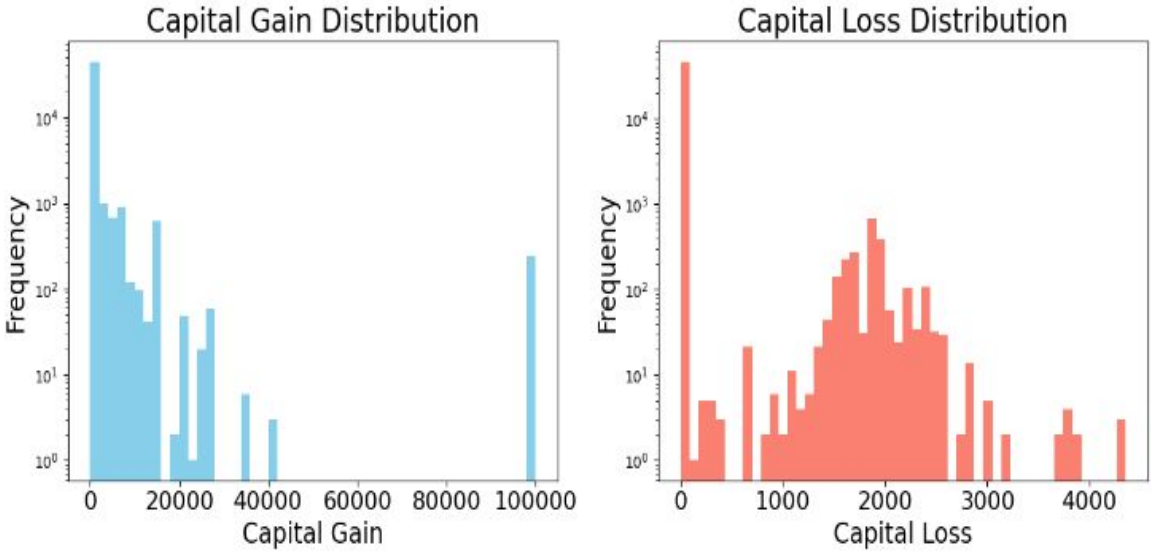
One-Hot Encoding for the categorical features.
Dataset went from **14 to 89** features. Why?

Dropped education column because it was already label encoded as education_num.

Binary Encoding:
(>50k) = 1
(<=50k) = 0

Target
(>50k) or (<=50k)

4. Dataset Preprocessing



Capital Gain/Loss: were binned and encoded into 3 categories:

- none
- high
- low

Hours-per-week: were binned and encoded into 3 categories:

- part-time
- full-time
- overtime

Two versions of the dataset:

df1 does **not** contain binning.

df2 does contain binning.

5. Splitting & Sampling

Three sampling approaches were chosen:

- **Raw splits (R):** No resampling applied.
- **Oversampling (O):** Replication of minorityclass samples using **RandomOverSampler.**
- **Undersampling (U):** Reduction of majority class samples using **RandomUnderSampler.**

Dataset Shape (48843, 14)

Example Test Split Sizes

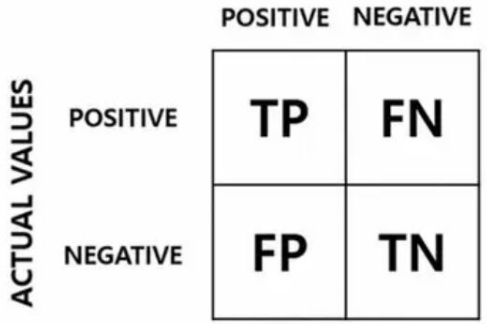
Split Type	df1 (Original)	df2 (Binned)	Features
Raw Split (Train)	39,073	39,073	89 / 95
Oversampled (Train)	59,448	59,448	89 / 95
Undersampled (Train)	18,698	18,698	89 / 95

Ten random splits per dataset version (df1 and df2) **and per sampling approach.**

Dataset
df1_O
df1_U
df1_R
df2_O
df2_U
df2_R

Splits were **80/20** with **stratification** to insure class imbalance preservation.

6. Model Training & Testing



$$Precision = \frac{TP}{TP + FP} \qquad Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Accuracy: Proportion of total correct predictions.
Precision: Out of all predicted positives, how many are actually positive.
Recall: Out of all actual positives, how many were correctly predicted.
F1-Score: Harmonic mean of precision and recall. Best when you want a balance.
ROC-AUC: Model's ability to rank positive vs. negative across thresholds.

Logistic Regression: Simple for binary classification. With StandardScaler().
Random Forest Classifier: A powerful ensemble method using decision trees.
Gradient Boosting (XGBoost): High-performance boosting algorithm that captures complex patterns and often yields state-of-the-art results.

7. Model Performance Summary

Dataset	Accuracy	Precision	Recall	F1 Score	ROC AUC
df1_O	0.81 ± 0.00	0.56 ± 0.01	0.84 ± 0.01	0.67 ± 0.01	0.90 ± 0.00
df1_U	0.80 ± 0.00	0.56 ± 0.01	0.84 ± 0.01	0.67 ± 0.01	0.90 ± 0.00
df1_R	0.85 ± 0.00	0.73 ± 0.01	0.60 ± 0.01	0.66 ± 0.01	0.90 ± 0.00
df2_O	0.81 ± 0.00	0.57 ± 0.01	0.84 ± 0.01	0.68 ± 0.01	0.91 ± 0.00
df2_U	0.81 ± 0.00	0.57 ± 0.01	0.85 ± 0.01	0.68 ± 0.01	0.91 ± 0.00
df2_R	0.85 ± 0.00	0.75 ± 0.01	0.59 ± 0.01	0.66 ± 0.01	0.91 ± 0.00

TABLE IV: Logistic Regression Performance (mean ± std)

Dataset	Accuracy	Precision	Recall	F1 Score	ROC AUC
df1_O	0.85 ± 0.00	0.68 ± 0.01	0.68 ± 0.01	0.68 ± 0.01	0.90 ± 0.00
df1_U	0.81 ± 0.00	0.58 ± 0.01	0.83 ± 0.01	0.68 ± 0.01	0.90 ± 0.00
df1_R	0.85 ± 0.00	0.73 ± 0.01	0.62 ± 0.01	0.67 ± 0.01	0.90 ± 0.00
df2_O	0.84 ± 0.00	0.65 ± 0.00	0.66 ± 0.01	0.66 ± 0.00	0.89 ± 0.00
df2_U	0.81 ± 0.00	0.57 ± 0.01	0.81 ± 0.01	0.67 ± 0.00	0.90 ± 0.00
df2_R	0.84 ± 0.00	0.70 ± 0.00	0.60 ± 0.01	0.65 ± 0.01	0.89 ± 0.00

TABLE V: Random Forest Performance (mean ± std)

Dataset	Accuracy	Precision	Recall	F1 Score	ROC AUC
df1_O	0.84 ± 0.00	0.61 ± 0.01	0.85 ± 0.01	0.71 ± 0.00	0.93 ± 0.00
df1_U	0.83 ± 0.00	0.60 ± 0.01	0.87 ± 0.01	0.71 ± 0.00	0.93 ± 0.00
df1_R	0.87 ± 0.00	0.78 ± 0.01	0.65 ± 0.01	0.71 ± 0.01	0.93 ± 0.00
df2_O	0.82 ± 0.00	0.59 ± 0.01	0.83 ± 0.01	0.69 ± 0.00	0.91 ± 0.00
df2_U	0.81 ± 0.00	0.57 ± 0.01	0.85 ± 0.01	0.68 ± 0.00	0.91 ± 0.00
df2_R	0.86 ± 0.00	0.74 ± 0.01	0.62 ± 0.01	0.68 ± 0.01	0.91 ± 0.00

TABLE VI: XGBoost Performance (mean ± std)

Observations:

df1 vs df2: Results were the same, only negligible differences occur.
Great ROC-AUC Scores: Regardless of what strategy/model we choose roc-auc is always great.
Recall vs Precision vs F1-Score: We cannot have high recall and precision at the same time so f1-score is always "low" around 70%.
STD is always "null": showcases our models are not sensitive to how the data is split, they handle randomness well.
Sampling increases recall: but at what cost?
Precision. Overall sampling increases (>50k) predictions right and wrong.
Similar results all-across: to improve the results we'd need more and higher quality data.

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rparameter Tuning

Hyperparameter Tuning Methodology

Objective: Maximizing relative performance for each use case.
High Risk Loan: XGBoost Df1_R - Highest Precision
IRS Audit: XGBoost Df1_U - Highest Recall
Social Security Measures: XGBoost Df1_R - Highest Recall Class 0

GridSearchCV : A tool that tests different combinations of model settings to find the best one.

StratifiedKFold: Splits data so that each split has the same proportion of each class. 5 folds means trains on 4 folds and tests on 1 fold, rotating folds.

Output: Best combination of params. Models were then re-trained and re-tested with these params.

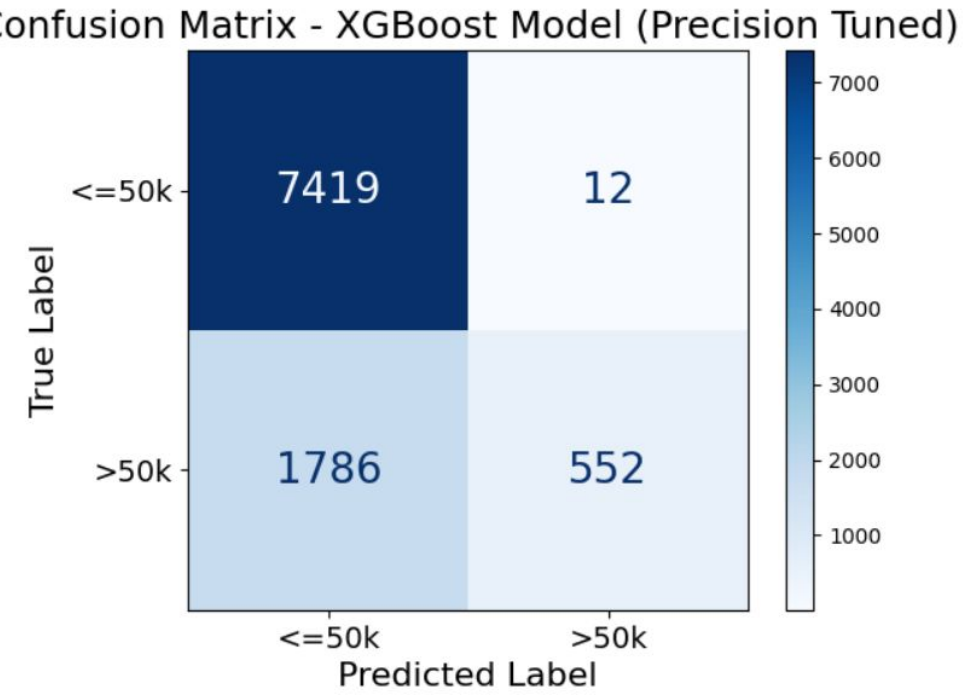
```
xgb_param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [3, 6, 10],
    'learning_rate': [0.01, 0.1, 0.2],
    'subsample': [0.8, 1.0],
    'colsample_bytree': [0.8, 1.0]
}
```

Tuning

1. Bank loan approval

Why: Banks want to *minimize* money lost and guarantee return in their operations.
Problem: They only have past and current information about their client and lack insight into what their life will look like in *future* years.

Solution: Focus on **precision** and lower **False Positives** .



Setting	Accuracy	Precision	Recall	F1 Score	ROC AUC
Before	0.87 ± 0.00	0.78 ± 0.01	0.65 ± 0.01	0.71 ± 0.01	0.93 ± 0.00
After	0.82 ± 0.00	0.98 ± 0.00	0.24 ± 0.01	0.38 ± 0.01	0.89 ± 0.00

True Negatives	False Positives	False Negatives	True Positives
7419	12	1786	552

$$\text{Precision}_1 = \frac{552}{552 + 12} \sim= 98\%$$

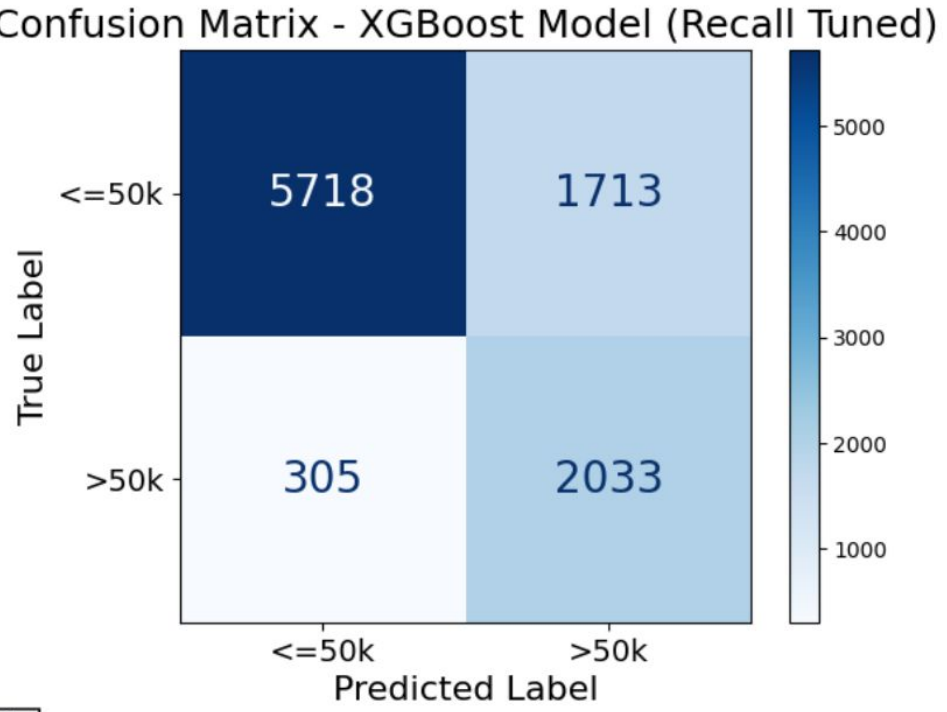
rparameter

Tuning

2. IRS auditing

Why: Some individuals try to forge their income to avoid paying taxes.
Problem: It's hard to find these individuals without taking a close look and spending time investigating.

Solution: Focus on **recall** to maximize the number of people found.



Setting	Accuracy	Precision	Recall	F1 Score	ROC AUC
Before	0.83 ± 0.00	0.60 ± 0.01	0.87 ± 0.01	0.71 ± 0.00	0.93 ± 0.00
After	0.78 ± 0.01	0.52 ± 0.01	0.88 ± 0.01	0.66 ± 0.01	0.90 ± 0.00

$$\text{Recall}_1 = \frac{2033}{2033 + 305} \sim= 88\%$$

True Negatives	False Positives	False Negatives	True Positives
5718	1713	305	2033

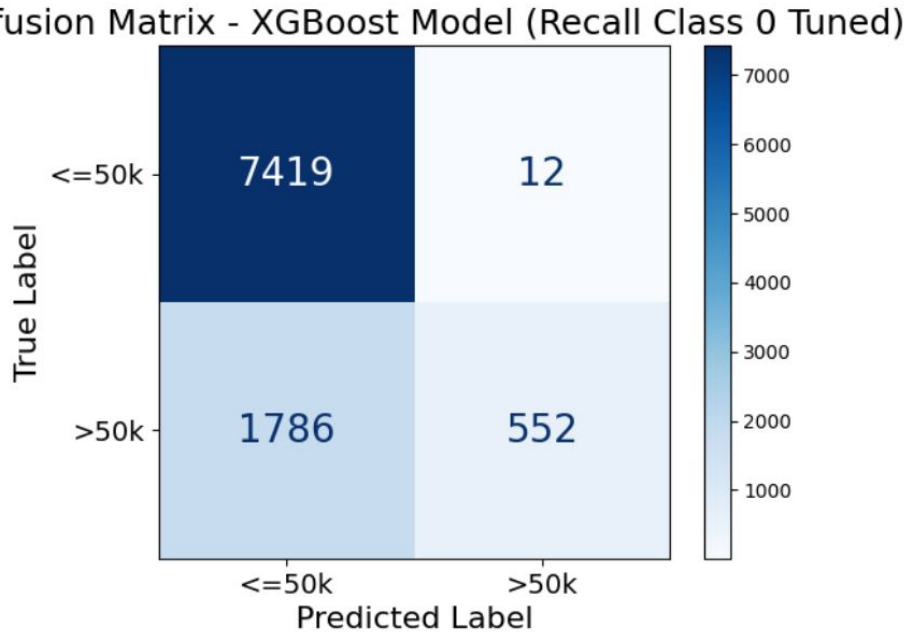
Tuning

3. Eligibility for Social Security benefits

Why: Institutions want to find all people that may need help and guarantee their well-being.

Problem: It's common for less fortunate people to be ashamed of their situation and not seek the help they need.

Solution: Focus on **recall** to reach everyone.



Setting	Accuracy	Precision	Recall	F1 Score	ROC AUC
Before	0.87 ± 0.00	0.90 ± 0.00	0.94 ± 0.00	0.92 ± 0.00	0.93 ± 0.00
After	0.82 ± 0.00	0.81 ± 0.00	1.00 ± 0.00	0.89 ± 0.00	0.89 ± 0.00

$$\text{Recall}_0 = \frac{7419}{7419 + 12} \sim= 100\%$$

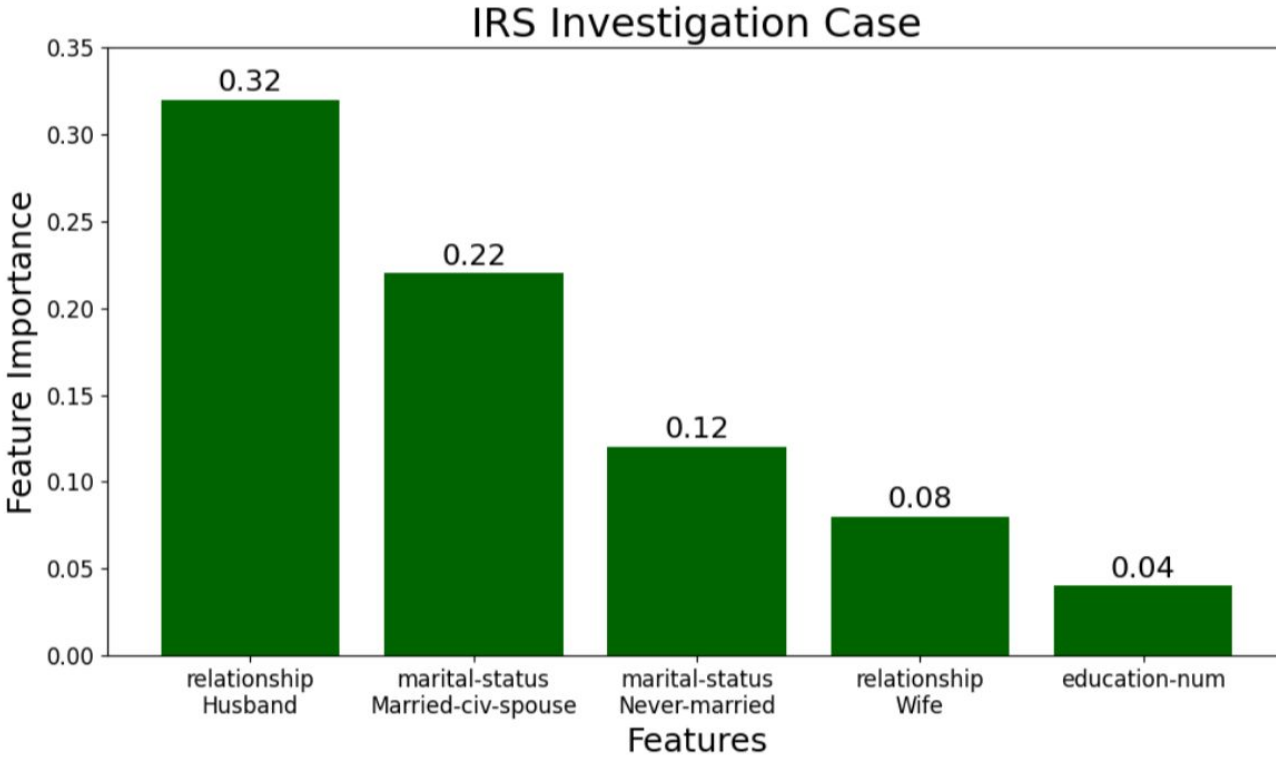
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7419	12	1786	552

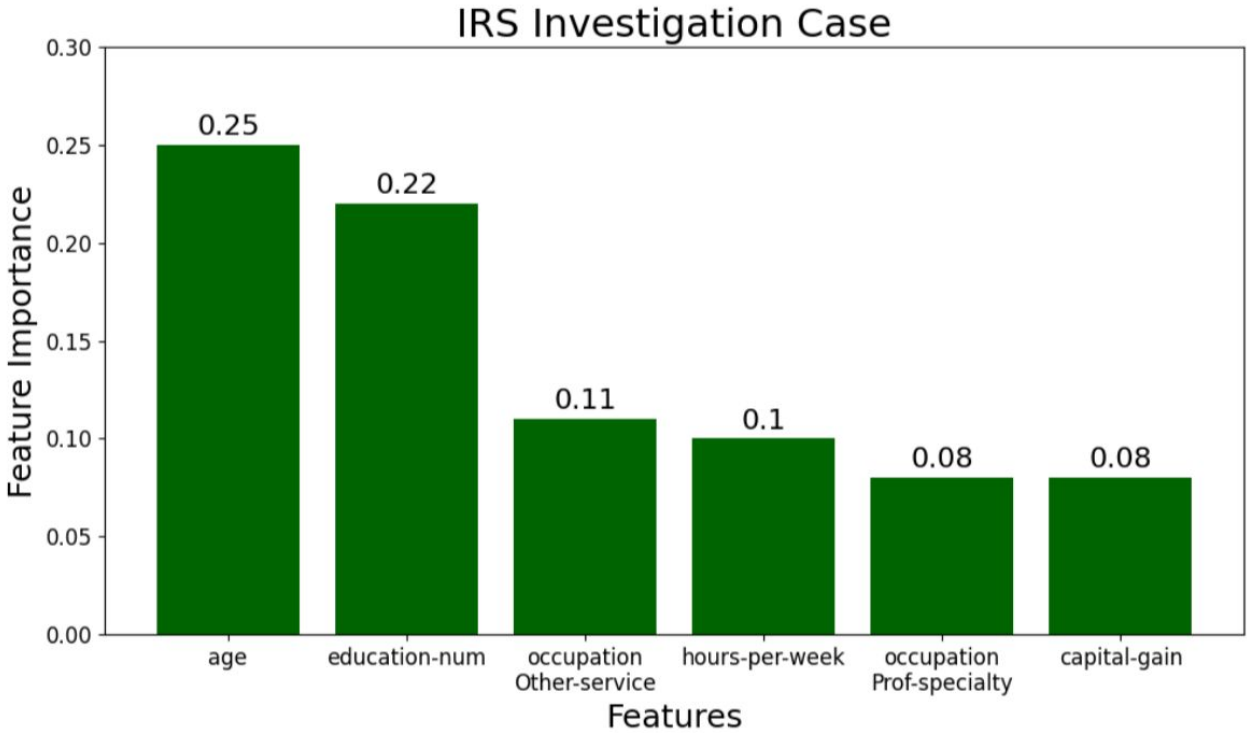
9. Fairness Analysis

EU AI Act - Recital 27: Privacy and data governance means that AI systems are developed and used in accordance with privacy and data protection rules, while processing data that meets high standards in terms of quality and integrity.

Features affected:

- race
- sex
- marital status
- native country
- relationship





Lessons Learned - Future Work & Conclusion

- **Collecting high quality data** is just as important or more than preprocessing it.
- **All options should be considered** when preprocessing/training.
- **Fairness** is a relevant issue that impacts model performance/bias.
- **Hyperparameter tuning** yielded great results for use case application.
- There are **different approaches** to ML we choose model value/applicability.

- Try **different datasets** from different parts of the world, different time periods.
- **Deploy** our models.

- We achieved models that **bring value** to their associated use case.
- On the other hand we also achieve **overall metrics** on par with kaggle 'better' examples. (87% acc & 93% roc-auc).

Predicting Income Levels Using Machine Learning: A Classification Approach with the Adult Census Dataset

Questions?

Scan the QR code to check our source code.



THANK YOU

Hugo Ribeiro - 113402

Pedro Ponte - 98059